Apurva Kumar

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Professional Summary

Electronics & Communication Engineering student with hands-on experience in embedded systems, computer architecture, and engineering research. Skilled in STM32/ESP32 embedded development, sensor interfacing, gem5-based system performance analysis, and technical problem solving. Passionate about applying engineering principles, data analytics, and digital technologies to enhance system reliability, operational efficiency, and engineering excellence.

Education

MIT World Peace University — Pune, India
B.Tech in Electronics & Communication Engineering (AI & ML Specialization) — *Expected June 2027 — CGPA: 8.58*

Technical Skills

Programming: C, C++, Python

Embedded Systems: Embedded C, STM32, ESP32, Arduino, STM32CubeMX, Keil, UART, SPI, I2C, Sensor Interfacing

Software & Tools: MATLAB, Simulink, Docker, Git, GitHub, Linux, VS Code, AMD Vivado, Cadence Virtuoso, Multisim

Computer Architecture: gem5, SPEC CPU2017, Cache Analysis, Memory Hierarchy, Network-on-Chip (NoC)

AI & Machine Learning: TensorFlow, PyTorch, Scikit-learn, OpenCV

Experience

Embedded Systems Intern — June 2026 – Present
Tata Technologies — *Pune, India*

- Contributed to embedded software development and validation for automotive electronic systems.
- Assisted in firmware integration, hardware testing, and engineering verification of embedded control modules.

Research Intern — May 2026 – June 2026
Indian Institute of Technology Guwahati — *Guwahati, India*

- Instrumented cache statistics and evaluated memory hierarchy performance using gem5 and SPEC CPU2017 benchmarks.
- Implemented and validated Network-on-Chip routing algorithms using C++, Python, and Docker.

Automotive Systems Trainee — June 2025
International Automobile Centre of Excellence (iACE) — *Gujarat, India*

- Completed hands-on training in automotive embedded systems, focusing on ECU architecture, sensor interfacing, CAN communication, and ADAS technologies.
- Gained practical exposure to electric vehicle systems, automotive diagnostics, and embedded software integration through laboratory-based training.

Research Publications

Noise-Conditioned Residual Learning for Robust Channel Estimation in Massive MIMO — *Status: Under Review*

- Proposed a ConvLSTM-based residual learning architecture for improving channel estimation under high-mobility 3GPP scenarios.

FleetEase: Simulation-Driven ADAS Validation Framework — *Status: Under Review*

- Developed a CARLA and RoadRunner-based simulation framework integrating multi-sensor fusion and adaptive TTC reasoning for ADAS validation.

Engineering Projects

Embedded Driver Assistance System — *Aug 2024 - Nov 2024*

- Designed and developed a real-time obstacle detection system using ESP32, ultrasonic, and IR sensors.
- Developed Embedded C firmware for sensor integration, testing, and debugging to improve system reliability.

FleetEase Fleet & Booking Management Platform — *Aug 2025 - Dec 2025*

- Developed a full-stack fleet management platform using React, Flutter, Node.js, Express, and PostgreSQL.
- Implemented secure authentication, vehicle tracking, and booking management functionalities.

AI-Enabled Gas Detection using PANI–β-$AgVO_3$/Chitosan Composite Sensor — *Jan 2026 - Present*

- Designed an embedded gas monitoring system integrating MiCS gas sensors for real-time environmental monitoring.
- Implemented intelligent data processing to enhance gas detection accuracy and system performance.

Computer Architecture Performance Analysis using gem5 — *May 2026*

- Performed cache hierarchy and AMAT performance analysis using gem5 and SPEC CPU2017 workloads.
- Evaluated memory hierarchy and Network-on-Chip (NoC) configurations using C++ to optimize system performance.

Certifications

Introduction to IoT – Coursera — Embedded C Programming – Coursera — Machine Learning with Python – IBM